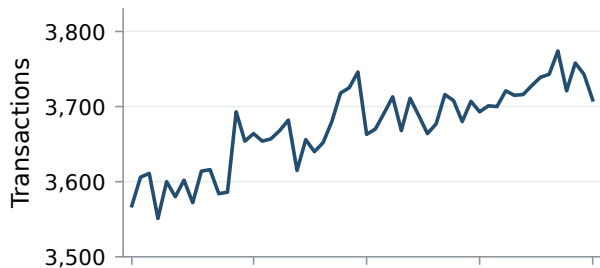
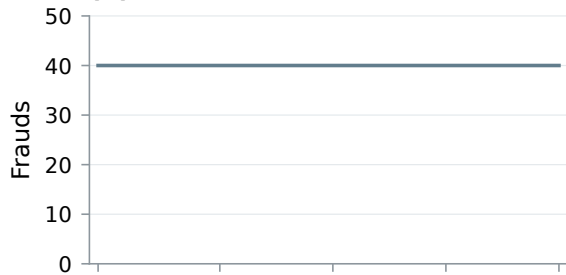
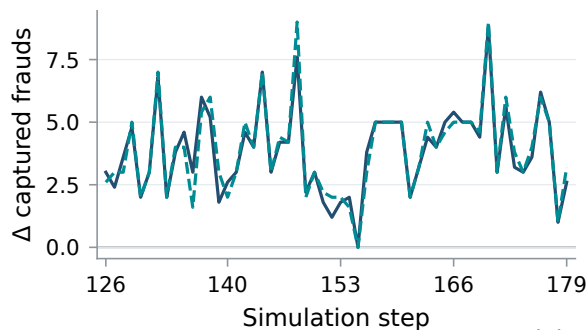
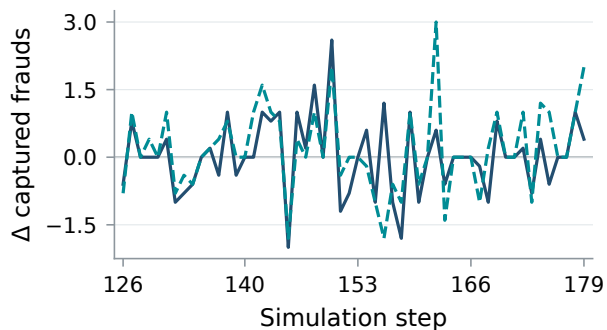


(a) Transaction volume**(b) Fraud count****(c) History above B_0** **(d) History above B_1** 

— LightGBM - - - XGBoost

Whole-window top-3% selections; no step-wise reranking.